

Housing Investment during the Pandemic

Muhebullah Karimzada, Shahed Khan and Terry Yip

McMaster U., Western U. and McMaster U.

CEA Conference 2022

Motivation

We are trying to understand what explains the housing investment in Canada during the Pandemic

- ▶ In March 2020, economic activities slowed down and unemployment spiked as a consequence of lockdown orders

▶ Unemployment rate

- ▶ According to the Labour Force Survey, working from home increased from 4% to 32%-40% during the Covid19 period

- ▶ Governments increased public debt and used a broad set of fiscal instruments to mitigate the crisis

- ▶ Bank of Canada used monetary policy to mitigate the crisis

▶ Bank rate

- ▶ However the housing market recovered very quickly during the crisis

▶ Home price rising

Motivation

Suggested explanations:

- ▶ Mortgage rates reduced significantly with the BoC's overnight rate cut to close to zero
▶ Home sales
- ▶ Government provided generous income support to households mostly affected by COVID-19 (such as CERB and other transfers)
- ▶ In 2020 Q2, Canadians received more money from government aid programs than they lost in wages and salaries

Other possible explanations:

- ▶ The supply of homes tightens
▶ Home supply tightens
- ▶ Working from homes is now reality for many Canadians
▶ Moving out

Outline of the model

The model environment closely follows Iacoviello and Neri (2010).

- ▶ Two types of households: patient and impatient
- ▶ Both types of HHs consume, work in two sectors (consumption sectors and housing sectors), and accumulate housing
- ▶ HHs can work at office or from home for consumption good sectors
- ▶ Unlike patient HHs, impatient HHs don't own capital or firms. Their borrowing limits are constrained by a fraction of the expected present value of their next period's housing stock.
- ▶ Other agents: firms, retailers and monetary policy authority

Patient Households

- ▶ The representative patient household maximizes life-time utility function of the form

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(\ln(c_t^p - \epsilon c_{t-1}^p) + j_t \ln h_t^p - \frac{\tau_t}{1+\eta} (n_{o,t}^{p1+\xi} + n_{wh,t}^{p1+\xi} + n_{h,t}^{p1+\xi})^{\frac{1+\eta}{1+\xi}} \right)$$

where, $n_{c,t}^p = (1 - \chi)n_{o,t}^p + \chi n_{wh,t}^{p\alpha_o} h_t^{p1-\alpha_o}$

- ▶ The budget constraint is

$$c_t^p + \frac{k_{c,t}^p}{A_{k,t}} + k_{h,t}^p + q_t h_t^p + s_t = w_{c,t} n_{c,t}^p + w_{h,t} n_{h,t}^p + (R_{c,t} + \frac{1 - \delta_{kc}}{A_{k,t}}) k_{c,t-1}^p + (R_{h,t} + 1 - \delta_{kh}) k_{h,t-1}^p + \frac{R_{t-1} s_{t-1}}{\pi_t} + q_t (1 - \delta_h) h_{t-1}^p \quad (1)$$

Impatient Households

- ▶ The impatient household maximizes the following:

$$E_0 \sum_{t=0}^{\infty} \beta^{it} \left(\ln(c_t^i - \epsilon^i c_{t-1}^i) + j_t \ln h_t^i - \frac{\tau_t}{1+\eta} (n_{o,t}^{i1+\xi} + n_{wh,t}^{i1+\xi} + n_{h,t}^{i1+\xi})^{\frac{1+\eta}{1+\xi}} \right)$$

where, $n_{c,t}^i = (1 - \chi)n_{o,t}^i + \chi n_{wh,t}^{i\alpha_o} h_t^{i1-\alpha_o}$

- ▶ The budget constraint is

$$c_t^i + q_t h_t^i + \frac{R_{t-1} b_{t-1}}{\pi_t} = w_{c,t} n_{c,t}^i + w_{h,t} n_{h,t}^i + q_t (1 - \delta_h) h_{t-1}^i + b_t \quad (2)$$

- ▶ The borrowing constraint is

$$b_t \leq m E_t \left(\frac{q_{t+1} h_{t+1}^i \pi_{t+1}}{R_t} \right) \quad (3)$$

Technology and Firms

- ▶ The firm hires labour and capital to produce wholesale goods Y_t and new houses IH_t . It maximizes the following:

$$\max \frac{Y_t}{X_t} + q_t IH_t - \left[\sum_{i=c,h} w_{i,t} n_{i,t}^p + \sum_{i=c,h} w_{i,t} n_{i,t}^i + \sum_{i=c,h} R_{i,t} k_{i,t-1} \right] \quad (4)$$

- ▶ The production technologies are:

$$Y_t = \left(A_{c,t} (n_{c,t}^{p\alpha} n_{c,t}^{i1-\alpha}) \right)^{1-\mu_c} (k_{c,t-1}^p)^{\mu_c} \quad (5)$$

$$IH_t = \left(A_{h,t} (n_{h,t}^p \alpha n_{h,t}^{i1-\alpha}) \right)^{1-\mu_h} (k_{h,t-1}^p)^{\mu_h} \quad (6)$$

Data and estimation

- ▶ The model is estimated for Canada using Bayesian estimation technique
- ▶ 11 observables: consumption, inflation, housing investment, capital investment, labor hours in consumption sector, labor hours in housing sector, house price index, bank rate, wages in consumption sector, wages in housing sector and GDP
- ▶ Time periods: 1997:Q1 to 2021:Q1
- ▶ 7 shocks: Prod. consumption sector, prod. housing sector, prod. capital sector, housing preference, labor supply, cost push and monetary policy

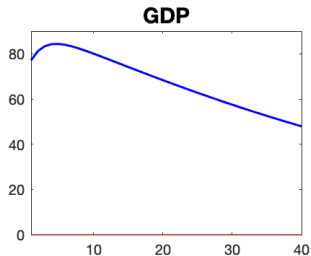
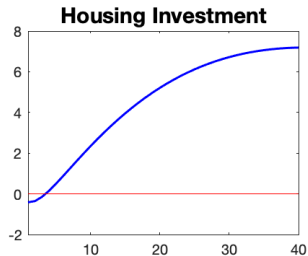
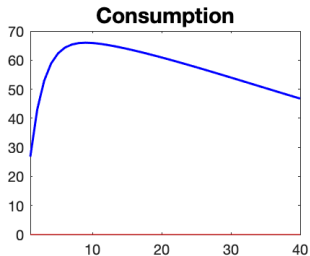
Data and estimation

We choose parameters (β , χ , j , m , and μ_c) to target the following data moment:

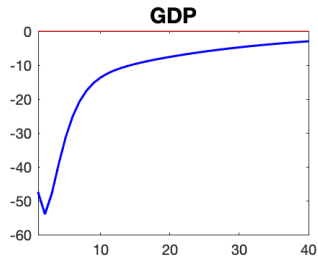
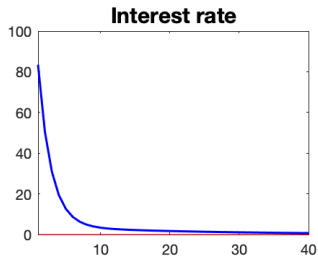
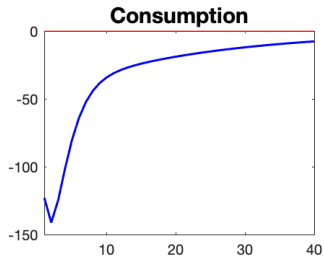
Table: Data Moments

Housing Investment to GDP	6%
Investment to GDP	11%
Working from Home Ratio	4%
Loan to Value Ratio	60%
Annual interest Rate	3%

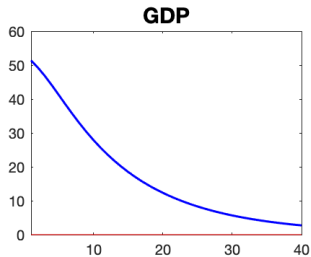
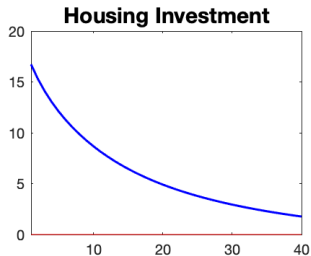
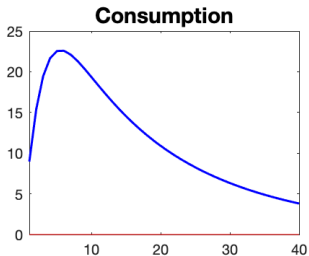
IRFs to Productivity Shock



IRFs to Monetary Shock

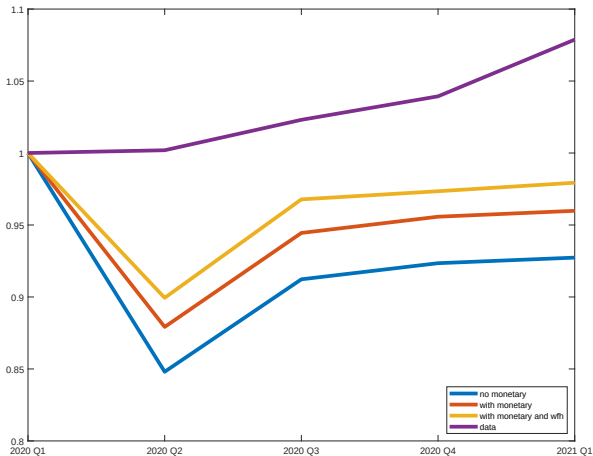


IRFs to working from home shock



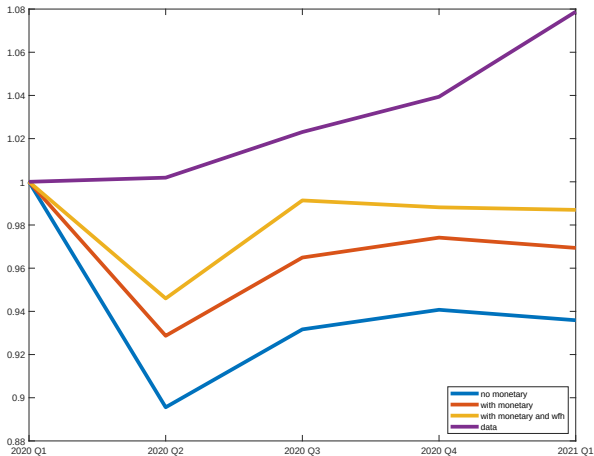
Simulated House Prices (Level) with 3 Different Scenarios

► Matching GDP



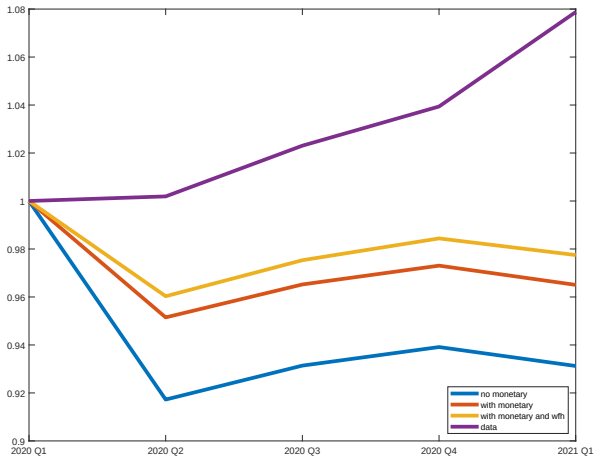
Simulated House Prices (Level) with 3 Different Scenarios

► Matching GDP and n_h

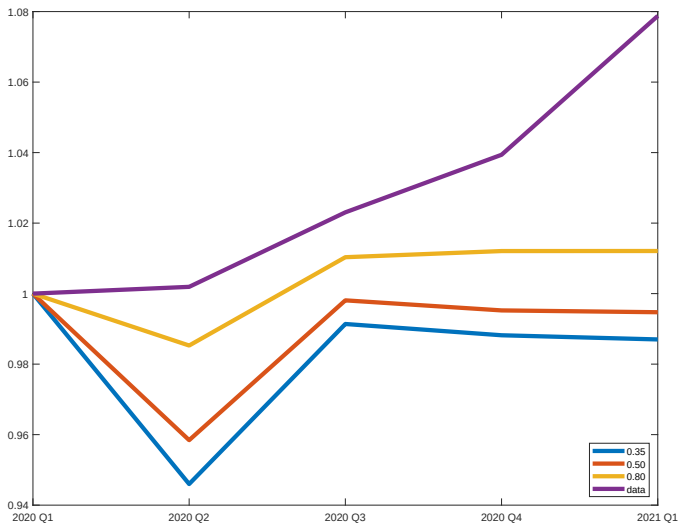


Simulated House Prices (Level) with 3 Different Scenarios

- Matching GDP, n_h and n_c



Simulated House Prices (Level) with 3 Different Scenarios



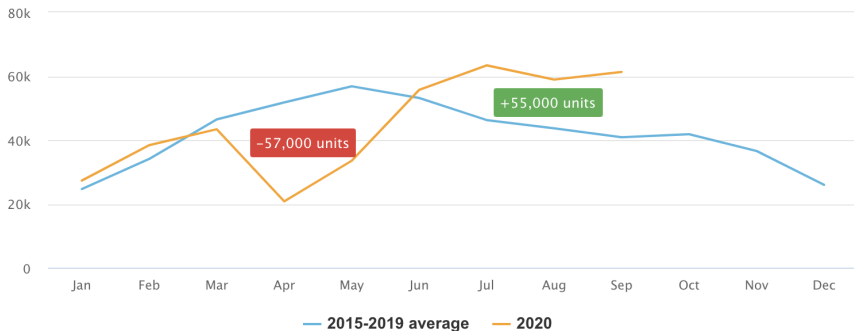
Conclusions

- ▶ We introduced working from home to the Iacoviello and Neri (2010) model and estimated it for Canada using Bayesian estimation technique
- ▶ Our preliminary results show that monetary policy shocks and working from home shocks are important in explaining house price movements during the Pandemic
- ▶ Future plan: adding government sector to the model to capture the effects of fiscal policy during the Pandemic

Home resales rebound quickly

Home resales soar back to life after spring lockdowns

Sales per month, unadjusted for seasonality



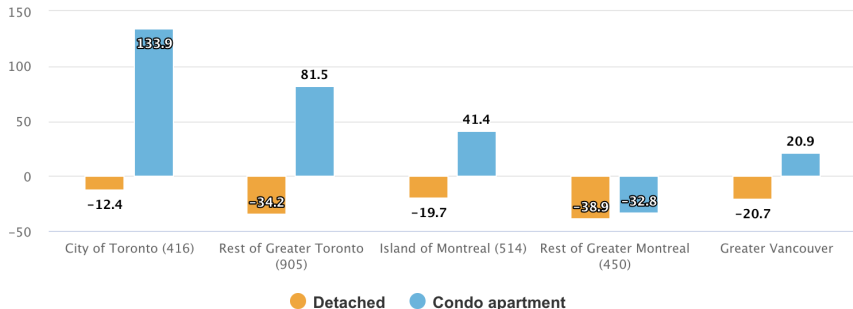
Source: Canadian Real Estate Association, RBC Economics

▶ back

Home supply tightens

Condo listings swell as supply of single family homes tightens

Annual % change, September 2020



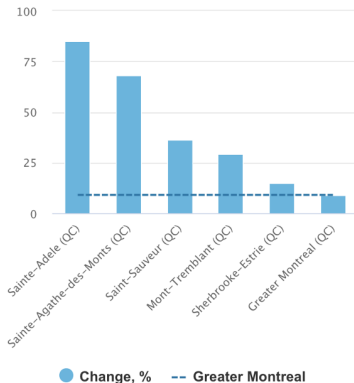
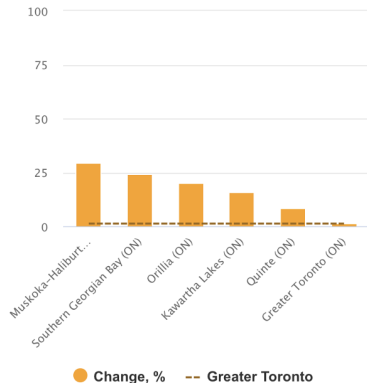
Source: Association professionnelle des courtiers immobiliers du Québec, Toronto Region Real Estate Board, Real Estate Board of Greater Vancouver, RBC Economics

[▶ back](#)

Moving out of big cities

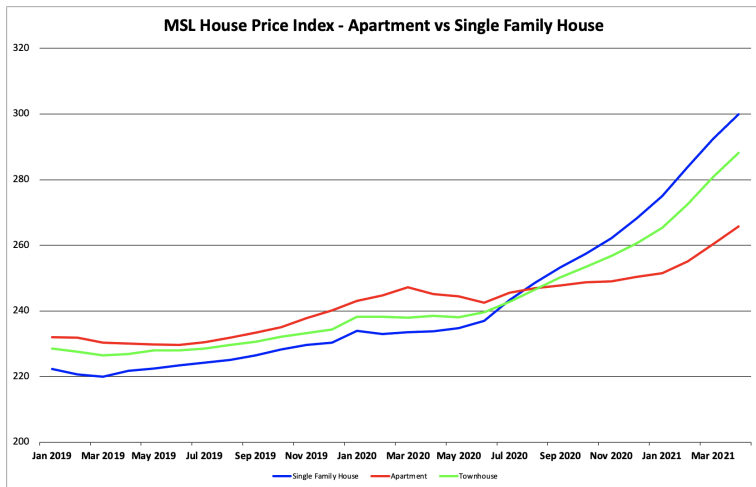
Cottage country home sales sizzle

Year-to-date, annual % change, third quarter 2020



Source: Association professionnelle des courtiers immobiliers du Québec, Canadian Real Estate Association, RBC Economics

Home price increasing



▶ back

Unemployment rate



▶ back

Overnight loan rate

